
5 Logic

Summarized version

A **statement** is a sentence that is definitely either true or false but not both. A **compound statement** is a combination of statements using the **logical connectives** or **operators**. These can be formally defined using **truth tables**, which give the truth value of a compound statement for every combination of truth values of its variables.

de Morgan's laws involve ands, ors, and nots. Specifically, using $\wedge$ to denote **and**, $\vee$ to denote **or**, and $\neg$ to denote **not**, one has that

$\neg(p \vee q)$ is logically equivalent to $\neg p \wedge \neg q$

$\neg(p \wedge q)$ is logically equivalent to $\neg p \vee \neg q$

A **conditional statement** or **implication** is a statement of the form “If p then q ”, denoted $p \Rightarrow q$. This is sometimes translated as “ p is sufficient for q ”, or “ $\neg p$ is necessary for $\neg q$ ”

It is useful to remember that:

- $p \Rightarrow q$ is logically equivalent to $\neg p \vee q$
- The **converse** of $p \Rightarrow q$ is $q \Rightarrow p$.
- The **contrapositive** of $p \Rightarrow q$ is $\neg q \Rightarrow \neg p$. The contrapositive is logically equivalent to the original.

A **biconditional statement** is a statement of the form “ p if and only if q ”, denoted $p \Leftrightarrow q$. This is sometimes translated as “ p is necessary and sufficient for q ”.